Software Cost & Effort Estimation Methodologies

RCS 305 ? Software Engineering | Riara University Revision Resource

1. The Estimation Challenge & Cone of Uncertainty

Early in the lifecycle, software cost estimates have high variance (+/- 400%). As requirements solidify through detailed design, the Cone of Uncertainty narrows toward exact projections.

2. Algorithmic Modeling: COCOMO II

Barry Boehm’s Constructive Cost Model (COCOMO) estimates Person-Months (PM) based on project size in Kilo Source Lines of Code (KSLOC):
Effort (PM) = A * (Size)^E * Product(Cost_Drivers)
Where Cost Drivers evaluate team capability, software complexity, database size, and tool maturity across Organic, Semidetached, and Embedded modes.

3. Non-Algorithmic Estimation

? Function Point Analysis (FPA): Measures functionality delivered based on External Inputs, Outputs, Inquiries, Internal Files, and External Interfaces.
? Expert Judgment & Planning Poker: Delphi and collaborative consensus techniques widely leveraged in Agile story pointing.

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